Unit I: Probability

Definitions of Probability, Set Operations,
Conditional Probability, Independence & Bayes' Theorem

STA201 – Theory of Probability and Statistics

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1 Introduction

Probability theory provides a mathematical framework to quantify uncertainty. For engineers, it is the language used to model:

- Reliability of systems (e.g., probability a component fails within a certain time),
- Quality control (e.g., probability a batch contains more than 2 defective items),
- Signal processing (e.g., probability of bit error in a communication channel),
- Risk analysis (e.g., probability of structural failure under extreme loads).

In this unit we build the set-theoretic foundations of probability and study the rules that govern the calculus of chance. Every example is chosen so that later we can attach a **random variable** (a numerical summary) to the outcomes, enabling us to compute expectations, variances, and full probability distributions (Unit II).

2 Random Experiments

A **random experiment** is a process that leads to one of several possible outcomes, where the outcome cannot be predicted with certainty beforehand. Examples: tossing a coin, rolling a die, measuring the lifetime of a battery, counting the number of defective items in a sample.

3 Sample Space

The **sample space** of a random experiment, denoted by S, is the set of all possible outcomes.

Examples:

- Tossing a coin: $S = \{H, T\}$.
- Tossing two coins: $S = \{HH, HT, TH, TT\}$.
- Rolling a die: $S = \{1, 2, 3, 4, 5, 6\}$.
- Lifetime of a component (in hours): $S = [0, \infty)$ (continuous).

4 Events

An **event** is any subset of the sample space S. We say the event *occurs* if the actual outcome of the experiment belongs to that subset.

Set Operations and Venn Diagrams

For events A and B in S:

- **Union** $A \cup B$: outcomes in A or B or both.
- **Intersection** $A \cap B$: outcomes in both A and B.
- **Complement** $A' = \bar{A} = S \setminus A$: outcomes not in A.
- Two events are **mutually exclusive** (disjoint) if $A \cap B = \varnothing$ (they cannot happen together).

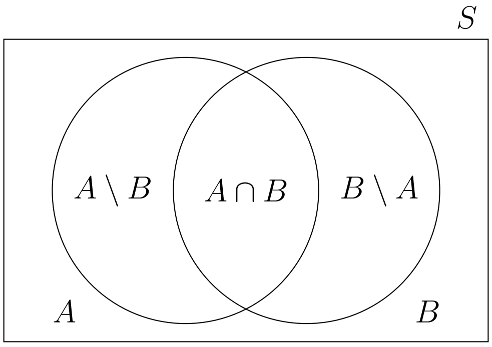

Figure 1: Venn diagram of events A and B in a sample space S.

5 Definitions of Probability

5.1 Classical Definition (Equally Likely Outcomes)

If the sample space S consists of a finite number N of outcomes that are **equally likely**, then for any event $A \subseteq S$,

$$\boxed{P(A) = \frac{|A|}{|S|} = \frac{\text{number of outcomes in } A}{\text{total number of outcomes}}.}$$

This definition assumes symmetry – fair coins, fair dice, well-shuffled cards, random selection from a batch.

5.2 Axiomatic Definition (Kolmogorov, 1933)

A probability measure P is a function that assigns to each event $A \subseteq S$ a real number such that:

1. **Non-negativity**: $P(A) \geq 0$ for every event A.
2. **Unit measure**: $P(S) = 1$.
3. **Countable additivity**: If $A_1, A_2, \ldots$ are mutually exclusive events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

From these axioms we derive:

- $P(\varnothing) = 0$.
- $P(A') = 1 - P(A)$.
- Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- If $A \subseteq B$, then $P(A) \leq P(B)$.

6 Set-Theoretic Probability Examples

Solved Example Tossing Two Coins

Two fair coins are tossed. Sample space:

$$S = \{HH, HT, TH, TT\}, \quad |S| = 4.$$

Define events:

- A: both coins show the same face. $A = \{HH, TT\}$.
- B: at least one head occurs. $B = \{HH, HT, TH\}$.
- C: exactly one head occurs. $C = \{HT, TH\}$.

Find:

(a) $P(A)$, $P(B)$, $P(C)$
(b) $P(A \cap B)$ (same face and at least one head)
(c) $P(A \cup B)$
(d) $P(A' \cap B)$ (not same face and at least one head)

Solution:

(a) $P(A) = 2/4 = 1/2$, $P(B) = 3/4$, $P(C) = 2/4 = 1/2$.
(b) $A \cap B = \{HH\}$, $P(A \cap B) = 1/4$.
(c) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{3}{4} - \frac{1}{4} = 1$ (indeed $A \cup B = S$).
(d) $A' = \{HT, TH\}$, $A' \cap B = \{HT, TH\} = C$, $P = 1/2$.

Solved Example Rolling a Fair Die

A fair six-sided die is rolled. $S = \{1, 2, 3, 4, 5, 6\}$, $|S| = 6$. Define:

- A: even number, $A = \{2, 4, 6\}$.
- B: number > 3, $B = \{4, 5, 6\}$.
- C: prime number, $C = \{2, 3, 5\}$.

Compute:

(a) $P(A)$, $P(B)$, $P(C)$
(b) $P(A \cap B)$ (even and > 3)
(c) $P(A \cup B)$
(d) $P(A' \cap C)$ (odd and prime)

Solution:

(a) $P(A) = 3/6 = 1/2$, $P(B) = 3/6 = 1/2$, $P(C) = 3/6 = 1/2$.
(b) $A \cap B = \{4, 6\}$, $P = 2/6 = 1/3$.
(c) $A \cup B = \{2, 4, 5, 6\}$, $P = 4/6 = 2/3$. (Addition: $\frac{1}{2} + \frac{1}{2} - \frac{1}{3} = 1 - \frac{1}{3} = 2/3$.)
(d) $A' = \{1, 3, 5\}$, $A' \cap C = \{3, 5\}$, $P = 2/6 = 1/3$.

Solved Example Drawing Cards

One card from a well-shuffled standard 52-card deck. Define:

- H: Heart (13 cards),
- K: King (4 cards),
- F: face card (J, Q, K; 12 cards).

Find:

(a) $P(H)$, $P(K)$, $P(F)$
(b) $P(H \cap K)$ (King of Hearts)
(c) $P(H \cup K)$
(d) $P(F' \cap H)$ (Heart, not face)

Solution:

(a) $P(H) = 13/52 = 1/4$, $P(K) = 4/52 = 1/13$, $P(F) = 12/52 = 3/13$.
(b) $H \cap K$: King of Hearts $\Rightarrow 1/52$.
(c) $P(H \cup K) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$.
(d) F': non-face (40 cards). Heart non-face: Ace–10 of Hearts (10 cards). $P = 10/52 = 5/26$.

Solved Example Manufacturing Defects

1000 components: 50 have defect A, 80 have defect B, 20 have both. Random selection.

- A: component has defect A,
- B: component has defect B.

Compute:

(a) $P(A)$, $P(B)$, $P(A \cap B)$
(b) $P(A \cup B)$ (at least one defect)
(c) $P(A' \cap B')$ (neither defect)

Solution: $P(A) = 50/1000 = 0.05$, $P(B) = 0.08$, $P(A \cap B) = 0.02$.

(a) see above.
(b) $P(A \cup B) = 0.05 + 0.08 - 0.02 = 0.11$.
(c) $P(A' \cap B') = P((A \cup B)') = 1 - 0.11 = 0.89$.

Solved Example Two Dice

Two fair dice (red and blue). $S = \{(i, j) \mid 1 \le i, j \le 6\}$, $|S| = 36$. Define:

- E: sum is even,

- T: sum ≥ 8,
- D: doubles (both numbers equal).

Find:

(a) $P(E)$, $P(T)$, $P(D)$
(b) $P(E \cap D)$ (even sum and doubles)
(c) $P(T \cup D)$
(d) $P(E' \cap T)$ (odd sum and sum ≥ 8)

Solution (sketch):

(a) Even sums: 2,4,6,8,10,12 $\to$ 18 outcomes, $P(E) = 1/2$; sum ≥ 8: 15 outcomes, $P(T) = 5/12$; doubles: 6 outcomes, $P(D) = 1/6$.
(b) $E \cap D$: (2,2),(4,4),(6,6) $\to$ 3 outcomes, $P = 3/36 = 1/12$.
(c) $T \cap D$: (4,4),(5,5),(6,6) $\to$ 3 outcomes, so $P(T \cup D) = \frac{15}{36} + \frac{6}{36} - \frac{3}{36} = \frac{18}{36} = 1/2$.
(d) Odd sums $\geq$ 8: sums 9 (4 outcomes) and 11 (2 outcomes) $\to$ 6 outcomes, $P = 6/36 = 1/6$.

(Full count can be elaborated in class.)

7 Conditional Probability

If we know that event B has occurred, the sample space is reduced to B. The **conditional probability** of A given B is

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)}}, \qquad P(B) > 0.$$

This leads to the **multiplication rule**:

$$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A).$$

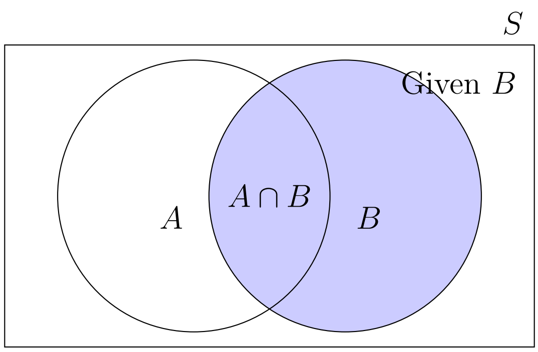

Figure 2: Conditional probability reduces the sample space to B.

Solved Examples on Conditional Probability

Solved Example Defects (revisited)

Using the data from Example 4 (1000 components: $P(A) = 0.05$, $P(B) = 0.08$, $P(A \cap B) = 0.02$), find the probability that a component has defect A given it has defect B.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.02}{0.08} = 0.25.$$

Likewise, $P(B \mid A) = \frac{0.02}{0.05} = 0.40$. So among A-defective parts, 40% also have B.

Solved Example Fuses

A lot of 100 fuses contains 10 defectives. Two fuses are drawn without replacement.

(a) What is the probability that the second fuse is defective given the first is defective?
(b) What is the probability that both are defective?

Solution: Let D_1: first defective, D_2: second defective.

(a) After drawing a defective first, 9 defectives remain out of 99, so $P(D_2 \mid D_1) = 9/99 = 1/11$.
(b) $P(D_1 \cap D_2) = P(D_1)P(D_2 \mid D_1) = \frac{10}{100} \cdot \frac{9}{99} = \frac{1}{110}$.

Solved Example Cards and Conditional

From a standard deck, one card is drawn. Let Q be the event "Queen" and H "Heart". Find $P(Q \mid H)$ and $P(H \mid Q)$.

$$P(Q) = 4/52 = 1/13, \; P(H) = 13/52 = 1/4, \; P(Q \cap H) = 1/52.$$

$$P(Q \mid H) = \frac{1/52}{13/52} = \frac{1}{13} = P(Q) \quad \text{(independence discussed later)},$$

$$P(H \mid Q) = \frac{1/52}{4/52} = \frac{1}{4} = P(H).$$

Solved Example Weather and Failure

A sensor fails with probability 0.1 under normal weather and 0.25 under extreme weather. The probability of extreme weather on any day is 0.05. If the sensor fails today, what is the probability that the weather is extreme?

$$P(\text{extreme} \mid \text{fail}) = \frac{P(\text{fail} \mid \text{extreme})P(\text{extreme})}{P(\text{fail})}.$$

First compute total failure probability:

$$P(\text{fail}) = 0.25 \cdot 0.05 + 0.1 \cdot 0.95 = 0.0125 + 0.095 = 0.1075.$$

Then

$$P(\text{extreme} \mid \text{fail}) = \frac{0.0125}{0.1075} \approx 0.1163.$$

(This also previews Bayes' theorem.)

8 Independence

Two events A and B are **independent** if the occurrence of one does not affect the probability of the other:

$$\boxed{P(A \cap B) = P(A)\,P(B)}.$$

Equivalently, $P(A \mid B) = P(A)$ (when $P(B) > 0$).

Note: Independence is not the same as mutual exclusivity. Mutually exclusive events with non-zero probability are always *dependent* because $P(A \mid B) = 0 \neq P(A)$.

Solved Examples on Independence

Solved Example Series System

A circuit has two independent components in series. The system works only if both work. If $p_1 = 0.9$ and $p_2 = 0.85$ are the probabilities each component works, the reliability is

$$P(\text{system works}) = P(\text{comp1 works} \cap \text{comp2 works}) = 0.9 \times 0.85 = 0.765.$$

Solved Example Parallel System

The same components in parallel: system works if at least one works.

$$P(\text{system works}) = 1 - P(\text{both fail}) = 1 - (1 - 0.9)(1 - 0.85) = 1 - 0.1 \times 0.15 = 0.985.$$

Solved Example Cards Independence Check

From a deck, events H (Heart) and K (King) are independent because

$$P(H \cap K) = 1/52, \quad P(H)P(K) = (13/52) \cdot (4/52) = 52/2704 = 1/52.$$

Solved Example Dice Independence

Roll two fair dice. Let A: first die shows 6, B: sum is odd. Determine if A and B are independent.

$$P(A) = 1/6, \quad P(B) = 18/36 = 1/2.$$

$A \cap B$: first die 6 and sum odd $\to$ second die must be odd (1,3,5) $\to$ 3 outcomes, $P(A \cap B) = 3/36 = 1/12$. Check: $P(A)P(B) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12} = P(A \cap B)$, so **independent**.

9 Bayes' Theorem

If $B_1, B_2, \ldots, B_k$ partition S (mutually exclusive, exhaustive), then for any event A with $P(A) > 0$,

$$\boxed{P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_{j=1}^{k} P(A \mid B_j)P(B_j)}}.$$

The denominator is the **law of total probability**: $P(A) = \sum_j P(A \mid B_j)P(B_j)$.

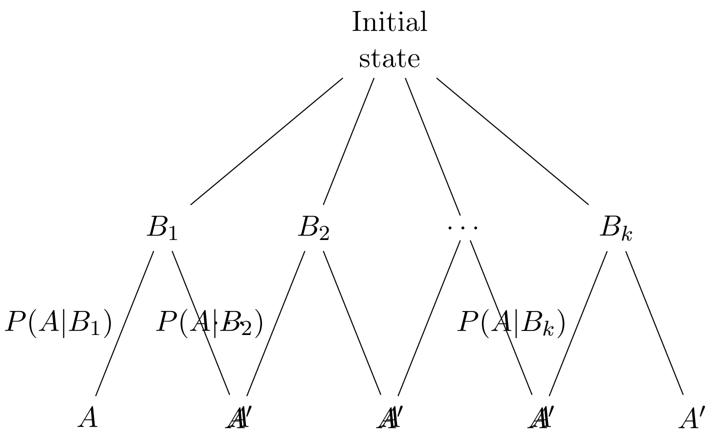

Figure 3: Tree diagram for total probability and Bayes' theorem.

Solved Examples on Bayes' Theorem

Solved Example Supplier Defects

Supplier X: 70% of parts, defect rate 2%. Supplier Y: 30%, defect rate 5%. Random part is defective. What is $P(Y \mid D)$?

$$P(X) = 0.7,\ P(D \mid X) = 0.02,$$
$$P(Y) = 0.3,\ P(D \mid Y) = 0.05.$$

Total probability:

$$P(D) = 0.7 \times 0.02 + 0.3 \times 0.05 = 0.014 + 0.015 = 0.029.$$

Bayes:

$$P(Y \mid D) = \frac{0.3 \times 0.05}{0.029} = \frac{0.015}{0.029} \approx 0.5172.$$

Even though Y supplies only 30%, it accounts for over half the defects.

Solved Example Jet Engine Fault

Diagnostic test: sensitivity 99%, specificity 97%. Base rate 0.5%. Given positive test, probability of actual fault?

$$P(F) = 0.005,\ P(+ \mid F) = 0.99,$$
$$P(+ \mid \bar{F}) = 1 - 0.97 = 0.03.$$

$$P(+) = 0.005 \times 0.99 + 0.995 \times 0.03 = 0.00495 + 0.02985 = 0.0348.$$

$$P(F \mid +) = \frac{0.00495}{0.0348} \approx 0.1422.$$

Despite high accuracy, a positive test only raises the probability to about 14%.

Solved Example Structural Flaw Detection

A non-destructive test for a structural flaw has sensitivity 95% and specificity 90%. The prior probability of a flaw is 2%. Find $P(\text{flaw} \mid \text{positive})$.

$$P(F) = 0.02,\ P(+ \mid F) = 0.95,$$
$$P(+ \mid \bar{F}) = 0.10.$$

Total positive probability:

$$P(+) = 0.02 \times 0.95 + 0.98 \times 0.10 = 0.019 + 0.098 = 0.117.$$

$$P(F \mid +) = \frac{0.019}{0.117} \approx 0.1624.$$

Again, a positive result is far from certain.

10 From Events to Random Variables (Preview)

Many of our examples count occurrences or flag conditions. In Unit II, we will define a **random variable** X mapping outcomes to numbers:

- Two-coin toss: X = number of heads; $P(X = 1) = P(\{HT, TH\}) = 1/2$.
- Die: Y = face value; $P(Y \text{ even}) = P(\{2, 4, 6\})$.
- Defect inspection: $Z = 1$ if defect A present, else 0; a Bernoulli random variable with $p = 0.05$.

All probability rules (union, intersection, complement, conditional, independence) will be applied directly to events expressed in terms of random variables.

Key Formulas Summary

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A') = 1 - P(A)$
- $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$
- $P(A \cap B) = P(A \mid B)P(B)$
- Independence: $P(A \cap B) = P(A)P(B)$
- Total probability: $P(A) = \sum_i P(A \mid B_i)P(B_i)$
- Bayes: $P(B_i \mid A) = \dfrac{P(A \mid B_i)P(B_i)}{\sum_j P(A \mid B_j)P(B_j)}$

Practice Problems

1. Three fair coins are tossed. Write the sample space. Let A: at least two heads; B: the first coin is head. Compute $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \cup B)$, and $P(B \mid A)$.

2. A card is drawn from a well-shuffled deck. Let A be "face card" and B be "Spade". Compute $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \cup B)$, and verify if A and B are independent.

3. In a group of 200 students, 120 study Mathematics, 90 study Physics, and 50 study both. Find the probability that a randomly selected student studies (i) neither, (ii) only Mathematics, (iii) only one subject.

4. Two dice are rolled. Find the probability that the number on the red die is strictly greater than that on the blue die.

5. A circuit has two independent switches in series. Each switch has probability 0.02 of being open. What is the probability the circuit allows current?

6. In the supplier defect example, compute $P(X \mid D)$ and interpret.

7. A diagnostic test for a software bug has sensitivity 98% and specificity 96%. If 1% of modules contain a bug, what is the probability a module flagged by the test actually has a bug?

8. A manufacturer uses three machines that produce 40%, 35%, and 25% of parts, with defect rates 1%, 2%, and 3%, respectively. A randomly selected part is defective; what is the probability it came from the third machine?